

Configure a Static IPv6 Address

Example (Windows):

1. Open Command Prompt as Administrator.
2. Open Network Settings and select the primary network adapter.
3. Assign:
IPv6 Address: 2001:db8:1::10
Prefix Length: 64
4. Open Command Prompt and run:
ipconfig
5. Verify that the IPv6 address appears in the output.

Configure a Unique Local IPv6 Address

Example:

IPv6 Address: fd00:1234:1::10/64

Verification:

1. Configure another device with:
fd00:1234:1::20/64
2. Open Command Prompt/Terminal.
3. Run:
ping -6 fd00:1234:1::20
4. Successful replies confirm connectivity.

Temporary IPv6 Address Test

Steps:

1. Note the current temporary IPv6 address.
2. Disconnect and reconnect the network (or reboot).
3. Run:
`ipconfig`
or
`ip -6 addr`
4. Compare the temporary IPv6 address before and after reconnecting.

Observed Behavior:

The temporary IPv6 address changed after reconnecting to the network. This helps improve privacy because the device does not always use the same IPv6 address.

Zomato-Style Restaurant Search Script (Python)

```
``python
    restaurants = {
        "Pizza Hub": "2001:db8:1::101",
        "Burger Point": "2001:db8:1::102",
        "Food Palace": "2001:db8:1::103"
    }

search = input("Enter restaurant name: ").lower()

for name, ip in restaurants.items():
    if search in name.lower():
        print(name, "-", ip)
``
```

AI-Generated IPv6 Subnetting Scenario

Network: 2001:db8:1000::/48

Subnet 1 (Delivery Partners):
2001:db8:1000:1::/64

Subnet 2 (Restaurants):
2001:db8:1000:2::/64

Subnet 3 (Customers):
2001:db8:1000:3::/64

Manual IPv6 Configuration

Assigned Address:
2001:db8:1000:2::10/64

Steps:

1. Open network adapter settings.
2. Select IPv6 properties.

3. Enter:

IPv6 Address: 2001:db8:1000:2::10

Prefix Length: 64

4. Save settings.

5. Verify using:

ipconfig

or

ip -6 addr